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Self-organized criticality in the Abelian sandpile

Driving, dissipation, and avalanche statistics

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Abstract

Adding one grain to a sandpile can leave it stable or trigger an avalanche across many sites. We use the two-dimensional Bak–Tang–Wiesenfeld model to examine how these responses depend on input location, dissipative boundaries and event grouping. The primary simulations comprised 10,000,000 measured grain additions across the baseline and three extensions, with five runs per condition. Four-grain batches used the same inputs as their sequential controls. At $L = 64$, the uniform open model had mean height 2.10361 and mean topplings 153.033 per grain, compared with the finite-grid prediction 153.043. Central input bias and cylindrical boundaries increased mean topplings to 266.669 and 357.511, while mean loss remained close to one per grain. Separate, longer confirmation runs gave height differences between biased and uniform input compatible with zero. All 500,000 matched batches reproduced grouped sequential topplings and loss exactly. Their larger event sizes reflect the grouping of successive responses. Normalized discrete fits favour cutoff descriptions over pure powers on the finite ranges examined. Together, the comparisons relate avalanche statistics to the stationary configuration measure, the response to input and the definition of an event.

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Chapter 1

Introduction

One grain can settle on a stable sandpile or start a cascade of topplings, depending on the configuration it encounters. This variation makes the sandpile a useful model for studying how a fixed local rule produces responses over a wide range of sizes.

Bak, Tang, and Wiesenfeld introduced self-organized criticality (SOC) to describe systems that develop critical behaviour under slow driving, without an externally tuned critical parameter [1, 2]. Their sandpile model combines gradual input with rapid redistribution and loss at the boundary.

We ask how the stationary configuration measure, the response to external driving and the grouping of events each affect avalanche statistics. For the two-dimensional BTW model, we first compare height, flux and avalanche distributions across lattice sizes. The Abelian property provides exact checks on stabilization and on the mean response to input.

Three controlled extensions isolate the roles of driving, geometry and event definition. We bias input towards the centre while keeping the graph fixed, replace the open square by a cylinder while retaining uniform input, and compare four-grain batches with the same grains added sequentially. All comparisons retain the threshold-four toppling rule. Matching the inputs in the batch comparison lets us distinguish changes caused by grouping from differences between random input sequences.

Chapter 2

Model, observables, and physical interpretation

2.1 Dynamics and observables

Let $\Lambda_L = \{0, \ldots, L-1\}^2$ be the lattice, with stable heights $h(x) \in \{0, 1, 2, 3\}$. We use the local rule in the project specification [7]. Starting from an empty grid, the primary baseline chooses a site independently and uniformly, adds one grain and lets the grid relax completely. A legal elementary toppling requires $h(x) \geq 4$, subtracts four grains and transfers one in each axial direction. Grains sent outside the lattice leave the system. The threshold remains four at edges and corners. The next grain is added only after stabilization, and external time counts input events.

Let $u_t(x)$ count elementary topplings at site x in event t. The sitewise toppling-count vector u_t is called the odometer. Let X_t be the set of input sites. We measure

$$S_t = \sum_x u_t(x), \qquad A_t = \sum_x \mathbf{1}_{\{u_t(x)>0\}}, \tag{2.1}$$

$$D_t = \sum_x [4 - d_G(x)]u_t(x), \qquad \ell_{X_t} = \begin{cases} 1 + \max\limits_{u_t(x)>0} \min\limits_{s \in X_t} d_G^{\mathrm{geo}}(x, s), & S_t > 0, \\ 0, & S_t = 0. \end{cases} \tag{2.2}$$

Here $d_G(x)$ is the number of internal directional neighbours and d_G^{geo} is graph distance. On the open square, distance is Manhattan distance. With one input site, reach is the greatest distance from that site to a toppled site, plus one. This gives a single-site avalanche reach one and measures spatial extent. For several input sites, each toppled site is measured from its nearest source, so the interpretation depends on the source set.

Means of S and D include all events, including $S = 0$; per-grain values divide the totals by the number of grains added. The main avalanche distributions and correlations condition on $S > 0$ and retain $D = 0$; positive-loss fits condition on $D > 0$. We report the zero-event probabilities separately. Mean height uses total mass after every event, while height frequencies are sampled every 128 added grains after stabilization, including after zero events. For batches, elapsed input is measured in grains as well as events. The distance from a batch to a dissipative boundary is the minimum over its input sites.

2.2 Scale invariance and the SOC mechanism

Scale invariance means that changing scale leaves the shape of a relation unchanged, apart from an overall factor. Suppose a positive function satisfies $f(cx) = g(c)f(x)$ for every $c, x > 0$. Rescaling first by a and then by b gives $g(ab) = g(a)g(b)$. Under continuity or suitable measurability, $g(c) = c^{-\alpha}$, giving

$$f(x) = f(x_0)(x/x_0)^{-\alpha} = Cx^{-\alpha}. \tag{2.3}$$

A probability distribution also needs normalizable support. For an unbounded discrete power tail with $\alpha > 1$, $\mathbb{P}(X \geq n) \sim C' n^{-(\alpha-1)}$, so the probability mass function (PMF) and complementary cumulative distribution function (CCDF) have different exponents. On a finite lattice, microscopic and system-size cutoffs limit the range over which scaling can hold approximately.

The separation between slow input and rapid relaxation is central to the BTW model. Local thresholds allow activity to spread, interior topplings conserve grains and the boundary removes them. These features motivate examining height, transport and avalanche distributions together. A power-law description requires a stated fitting range and distributional evidence beyond a straight segment on logarithmic axes [3].

2.3 Boundaries, steady states, and infinite volume

If m_t grains enter an event, total mass satisfies

$$M_t = M_{t-1} + m_t - D_t, \qquad \overline{D} = \overline{m} - \frac{M_T - M_0}{T}. \tag{2.4}$$

Open boundaries allow grains to leave as new grains arrive. Every site connects to a sink, and stable mass is bounded by $3L^2$. The endpoint term in Eq. (2.4) therefore vanishes over a long window, even without a mixing argument. Average loss can approach average input while other statistics still show transients. Without dissipation, continued input to a finite system eventually exceeds the capacity of any stable configuration.

Starting from an empty infinite lattice, any finite collection of grains can stabilize by spreading outward [9]. Continued input in a finite source region increases the total mass and expands the affected region. Stabilization on an occupied infinite background or an infinite-volume stationary measure requires additional assumptions.

We use *macroscopic steady state* to mean that selected bulk observables show no systematic drift. *Statistical stationarity* is stronger: the joint probability laws remain unchanged under shifts in time. In the physical sense used here, *equilibrium* has no sustained externally driven transport; detailed balance is a common criterion in stochastic models. The relevant description of the driven sandpile is a nonequilibrium stationary state. Grains continue to enter and leave even when the height statistics settle.

Chapter 3

Abelian structure and exact benchmarks

3.1 Matrix conditions and the threshold

Let B be the adjacency matrix of the internal lattice sites. With the handout's negative-diagonal convention [7],

$$\Delta = -4I + B, \qquad K = -\Delta = 4I - B, \qquad h \longmapsto h - Ke_x. \tag{3.1}$$

Nonnegative off-diagonal entries of Δ ensure that a toppling adds grains to other sites or leaves them unchanged. Symmetry describes the undirected transfers used here. The negative diagonal removes grains from the toppled site, while nonpositive row sums prevent mass creation. A strictly negative total row sum means that some mass can dissipate. For the open square, row sums are zero in the interior, -1 at non-corner edges and -2 at corners for $L \geq 2$, giving a total of $-4L$. The single-site case has row sum -4. The threshold is $k = -\Delta_{xx} = 4$.

Sink connectivity supplies the further condition needed for stabilization: every site must have a path to a sink. Dissipation elsewhere cannot stabilize a disconnected component with no such path. Both graphs used here meet this condition. Symmetry describes these graphs but is not required for every generalized Abelian sandpile. The cylinder wraps horizontally and has sinks only at its top and bottom; its total row sum is $-2L$ at the lattice sizes studied.

Two inconsistencies in the handout affect the local rule and its example [7, Eq. (6) and Fig. 3]. The threshold in its Equation (6) should be $h(x) \geq -\Delta_{xx}$. In its Figure 3, the upper-middle toppling omits a grain transferred to the centre. Applying the stated local rule gives

$$\begin{pmatrix} 1 & 3 & 3 \\ 2 & 4 & 2 \\ 2 & 1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 0 \\ 2 & 2 & 2 \end{pmatrix}, \quad (S, A, D, \ell) = (4, 4, 4, 3). \tag{3.2}$$

The input mass is 19 and the final mass is 15. The initial array already includes the added grain.

3.2 Termination and unique stabilization

The Abelian property means that choosing a different legal toppling order leaves the stabilized configuration unchanged. We first establish termination, then show that every complete legal sequence has the same toppling counts. The argument follows the finite-graph treatment [6, Theorem 2.3 and Exercise 2.6] and Dhar's Abelian structure [5].

Termination. Attach missing neighbours to an absorbing sink. For a real vector z,

$$z^\top K z = \sum_{\{x,y\} \text{ internal edge}} (z_x - z_y)^2 + \sum_x [4 - d_G(x)] z_x^2. \tag{3.3}$$

Sink connectivity makes this expression positive for nonzero z, so K is invertible. Let $Kw = \mathbf{1}$. If w had a nonpositive minimum at x, then $(Kw)_x \leq [4 - d_G(x)]w_x \leq 0$, contradicting $(Kw)_x = 1$. Hence $w > 0$. The potential $\Phi(h) = w^\top h$ is nonnegative for nonnegative heights, and each elementary toppling changes it by

$$\Phi(h - Ke_x) - \Phi(h) = -w^\top K e_x = -1. \tag{3.4}$$

The potential drops by one at each toppling and stays nonnegative. Only finitely many topplings are therefore possible, and continuing while an unstable site exists must reach a stable configuration. This applies to both the open graph and the dissipative cylinder.

Local commutation. Suppose distinct sites x, y are both unstable. Toppling either site leaves the other unstable, so both orders are legal and

$$T_y T_x h = h - Ke_x - Ke_y = T_x T_y h. \tag{3.5}$$

Unique stabilization. Let u, v count topplings in two complete legal stabilizations of the same configuration h. Follow the first sequence to the proposed toppling that would first exceed v at a site x. Just before it, the partial counts satisfy $a_x = v_x$ and $a_y \le v_y$ at every other site. Consequently,

$$(h - Ka)_x = h_x - 4v_x + \sum_y B_{xy} a_y \le h_x - 4v_x + \sum_y B_{xy} v_y = (h - Kv)_x \le 3. \tag{3.6}$$

Site x would already be stable, contradicting the legality of the proposed toppling. Hence $u \le v$. Reversing the argument gives $u = v$, so both sequences have the same final configuration. Every observable determined by the odometer is therefore independent of the schedule.

3.3 Stationary measures, ergodicity, and mean response

On a finite graph connected to a sink, positive addition probability at every site makes the dynamics eventually enter a single recurrent class with probability one. Its stable configurations can reach one another through additions, and subsequent additions cannot leave the class. Stable configurations outside it are transient. Each addition permutes the recurrent class, and any probability mixture of the permutations preserves its uniform measure [6, Theorems 2.9, 2.11; Exercise 2.16]. Changing the full-support driving distribution on the *same graph* therefore preserves stationary height statistics. This measure is uniform over recurrent configurations; individual-site heights are correlated and their frequencies can be unequal.

Time averages are ergodic on this finite recurrent class. Convergence of the distribution at a fixed time also requires aperiodicity. The one-site model, for example, cycles through four heights; sampling every four additions preserves its initial phase. This is why we compare time and ensemble estimates at specified sampling ages and use the recurrent-state theory for the ergodic statement.

Assigning equal probabilities to the four stable heights gives a reference mean of $3/2$. The literature value $17/8$ is the infinite-volume recurrent mean [8]. We compare both with the measured height frequencies in Section 5.1.2.

Finite grids also have an exact benchmark for the mean response. Let a_t record the grains added to each site in event t, counting repeated inputs, m be the number of grains per event in that condition, and q be the normalized input-site probability vector. The configuration balance and its stationary expectation give

$$h_t = h_{t-1} + a_t - Ku_t, \tag{3.7}$$

$$K\mathbb{E}[u_t] = \mathbb{E}[a_t] = mq, \qquad \mathbb{E}[S_t] = \mathbf{1}^\top K^{-1}(mq), \qquad \mathbb{E}[D_t] = m. \tag{3.8}$$

Equation (3.8) is the weighted form of Dhar's mean-toppling formula [6, Theorem 2.18]. We evaluate it by solving sparse linear systems and checking their residuals. For any finite input sequence, the potential used in the termination proof gives

$$\sum_t S_t = \sum_t w^\top a_t + w^\top (h_0 - h_T), \qquad Kw = \mathbf{1}. \tag{3.9}$$

The endpoint correction is $\Phi(h_0) - \Phi(h_T)$, where $\Phi(h) = w^\top h$. This identity provides an exact check for a finite input sequence.

Chapter 4

Computation and experimental design

4.1 Independent implementations and validation

A reference simulation scans the grid and performs one elementary toppling at a time. The faster algorithm keeps a queue of unstable sites and performs $\lfloor h(x)/4 \rfloor$ consecutive legal topplings at each visit, adding that count to the odometer. Each site has at most one pending place in the queue. This avoids repeated scans of stable sites and duplicate queue entries. We compared both Python and C versions with the reference.

Every event was checked against the mass balance. We also checked actual grid mass using height frequencies every 128 added grains and direct sums during each run; a separate array-based simulation performed this check after every validation event. Heights and counts use signed 64-bit integers, and repeated batch locations receive every grain assigned to them.

The comparisons cover all $256 \times 4 = 1,024$ stable 2×2 additions and 1,000 bounded random configurations for each implementation. Forward, reverse and random legal toppling orders give identical results. Tests also cover rectangular and degenerate grids, repeated topplings, all four observables and the corrected handout example. Extension checks verify periodic connections, dissipation and distances on the cylinder; biased-input probabilities and frequencies; and batch odometer sums and area unions. Exact enumeration gives 192 recurrent 2×2 states and verifies the uniform invariant measure under uniform and unequal positive input probabilities, including four successive additions.

Known discrete distributions, sparse samples and autoregressive series were used to check the statistical calculations. For each fit, we compared several initial parameter choices and checked the best converged solution against an independent projected-gradient calculation. All 144 aggregate fits passed these checks.

4.2 Profiling and performance

Profiling identified repeated neighbour construction as a major cost, so we stored the neighbour lists. A later $L = 64$ profile placed most time in the active-site queue. We implemented this algorithm in C, with Python generating the input sites, and calculated observables as sites toppled. Table 4.1 compares stabilization and observable calculation for a fixed input sequence.

Implementation	Median (ms)	Range (ms)	Speed ratio
Elementary scan, Python	552.216	[531.345, 562.245]	1.0
Bulk queue, Python	107.030	[106.292, 107.353]	5.2
C queue, one event per call	95.477	[94.000, 107.416]	5.8
C queue, full input sequence	0.167	[0.166, 0.205]	3298.7

Table 4.1: Stabilization times for the same $L = 16$ calculation: 2000 predetermined inputs from an all-three grid, giving 25377 elementary topplings. Each implementation first completes one untimed run. Each of five timing samples then repeats complete calculations for at least 0.2 s, resetting the grid each time; reported times are per calculation. All methods give identical S, A, D, ℓ and final grids. Compilation (0.477 s) and initial loading/warm-up (0.751 s) are timed separately. Ranges are minimum to maximum; speed ratios use the Python scan median. Times exclude generating input sites and reading or writing data.

The Python queue is 5.2 times faster than the scan, combining active-site selection, consecutive topplings and stored neighbour lists. Processing the full input sequence in C further reduces the

overhead of calls between Python and C. The environment used Python 3.12.2, NumPy 1.26.4, SciPy 1.13.0, Matplotlib 3.8.4 and Apple Clang 17 (arm64 macOS, eight logical CPUs).

4.3 Simulation lengths and initial conditions

The preliminary runs used $L = 64$ for open uniform, open biased and cylindrical input, each starting from empty, random stable and all-three grids with independent random-number sequences. They followed $60L^2$ grains in the uniform and cylindrical cases and $100L^2$ under bias. A separate equilibrated, matched batch comparison checked 81,920 subsequent grains. Based on these runs, we used five seeds per condition, baseline sizes $L = 8, 16, 32, 64$, extension sizes $L = 32, 64$, and 200,000 measured grains per seed. Burn-in is $20L^2$ for baseline and cylinder, and $40L^2$ for bias. Batches start from the matched baseline's grid after burn-in.

The 50 primary runs contain 10,000,000 measured grains and 8,500,000 events, plus 2,080,000 burn-in grains. Each batch run has 50,000 events and uses the same initial grid and subsequent input sites as its single-grain control. Those two million batch grains form paired comparisons. Separate random-number sequences are used for the pilot, primary conditions apart from matched batches, toppling-order checks, the 100 independent $L = 32$ ensemble chains and block resampling. The later confirmation experiment in Section 5.4.1 uses additional seeds and grains. The primary simulations took 40.31 s, including 24.25 s in simulation calculations and 14.28 s in data storage. These times include burn-in.

4.4 Uncertainty and distribution fitting

Primary 95% Student-t intervals use five independent seed estimates, with four degrees of freedom. Intervals use the variation between run estimates, allowing events within a run to be correlated. Independent extension differences use Welch intervals; batch comparisons use exact pairing. The time-series comparison uses 24 complete contiguous blocks of 8,192 grains, covering 196,608 grains. The remaining grains are retained in all-event summaries. Block lengths of 4,096 and 16,384 grains and 200 block-bootstrap resamples assess sensitivity. Autocorrelation is calculated separately for height, topplings, loss, dissipation indicators and height frequencies. The initial-positive-pair correlation-time estimate uses a finite lag range and has a floor of one, which short-lag anticorrelation can reach.

Fits use the exact counts of raw integer observations, with normalized likelihood

$$p(x \mid \alpha, \lambda) = \frac{x^{-\alpha} e^{-\lambda x}}{\sum_{n=x_{\min}}^{x_{\max}} n^{-\alpha} e^{-\lambda n}}. \tag{4.1}$$

The candidates are power ($\lambda = 0$), cutoff power and exponential ($\alpha = 0$), fitted to identical observations and support. Numerical bounds are $-10 \le \alpha \le 10$ and $0 \le \lambda x_{\max} \le 10^5$; no fitted free parameter reaches them. Finite support allows negative effective slopes. Each seed must retain at least 1,000 observations. The fixed lower limits are $(16, 8, 1, 3)$ for (S, A, D, ℓ); sensitivity checks halve and double them, with a minimum of one. For S, the upper fitting limit L^2 selects an intermediate range. The other upper limits, L^2, $3L^2+1$ and $2L-1$, are physical bounds. We record observed maxima separately from these fitting limits.

Parameter intervals are calculated across the five seed fits. Kolmogorov–Smirnov (KS) distances measure discrepancies between fitted and empirical CDFs. KS and marginal log-score comparisons describe fit on the observed sample; the cutoff model has an extra parameter, and no IID goodness-of-fit p-values are assigned. Block sensitivity assesses the log-score comparison under temporal dependence. Logarithmically binned PMFs divide counts by bin width and sample count. Fitting uses the original integer observations, with each size and model analysed separately.

Chapter 5

Results

5.1 Stationarity, height, and sampling

5.1.1 Approach to the measured regime

The pilot trajectories show initial mass accumulation followed by fluctuations around similar mean heights (Figure 5.1). Empty starts give a lower mean in the first $10L^2$-grain window. The longest estimated height or frequency correlation time was about 1,195 grains, making the chosen block about seven times longer. These diagnostics guided burn-in and blocking. Each starting state had one independent pilot chain.

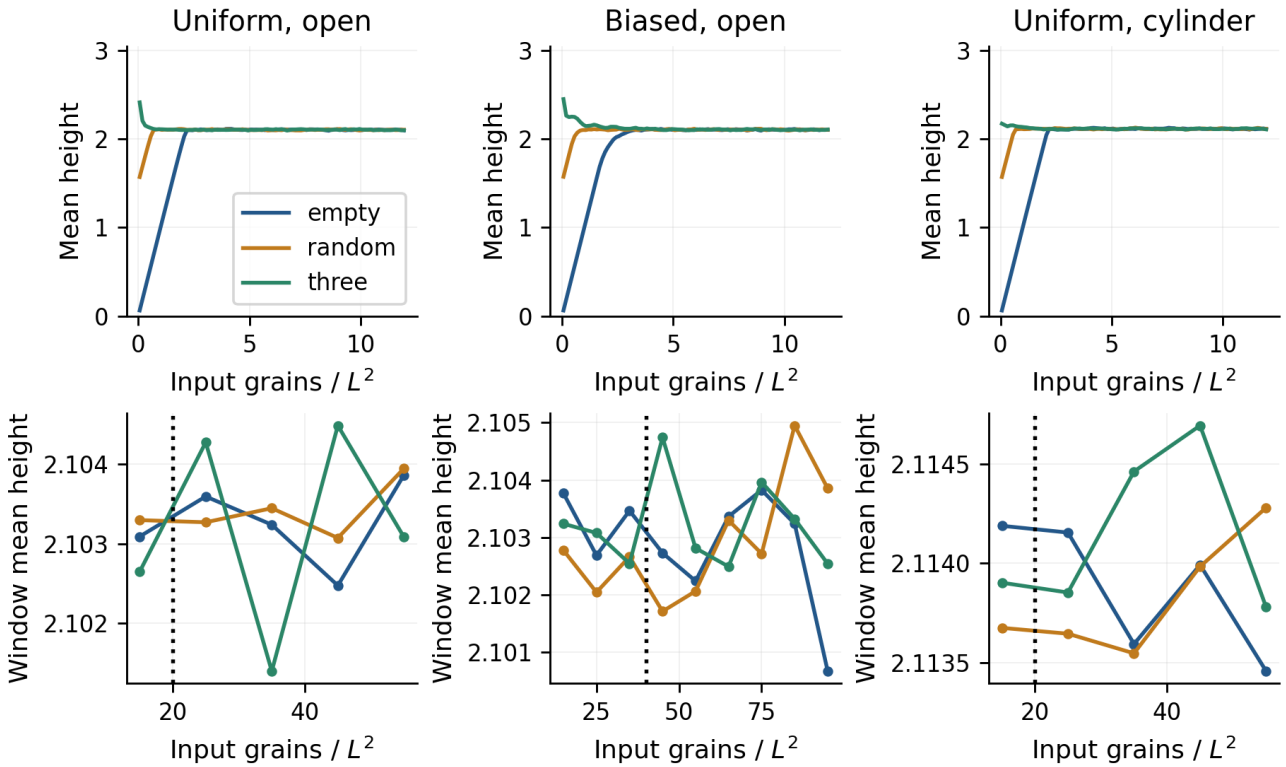


Figure 5.1: Different initial configurations approach similar late-stage height levels within each model, supporting the selected height burn-in. Independent $L = 64$ pilots start empty, random stable and all-three. Top: 512-grain mean heights over the first $12L^2$ grains. Bottom: connected $10L^2$-window means after the first window; vertical lines mark selected burn-in. Uniform/cylinder pilots last $60L^2$ grains; biased pilots last $100L^2$. Primary runs use separate seeds.

Across four consecutive measurement windows, height, all four height frequencies, mean topplings, loss and dissipation probability fluctuate without a common continuing drift. A small height difference under bias prompted the separate investigation in Section 5.4.1. Every primary run passed the mass and potential-balance checks in Eqs. (2.4) and (3.9), and every sparse benchmark residual was below 10^{-10}.

5.1.2 Height, finite size, and flux

Measured stable heights have unequal frequencies, with height three much more common than height zero (Figure 5.2). This weighting gives a mean above the equal-frequency reference 3/2.

Mean height rises from 1.98211 at $L = 8$ to 2.10361 at $L = 64$, approaching the infinite-volume value 17/8 (Table 5.1).

L	Mean height	Pred. S	Measured S	$P(S = 0)$	$P(D = 0 \mid S > 0)$	$P(D > 0)$
8	1.98211 ± 0.00114	3.462	3.463 ± 0.003	0.6194	0.2331	0.2919
16	2.04645 ± 0.00024	11.338	11.333 ± 0.018	0.5910	0.3883	0.2502
32	2.08355 ± 0.00044	40.580	40.611 ± 0.066	0.5743	0.5245	0.2024
64	2.10361 ± 0.00025	153.043	153.033 ± 0.426	0.5649	0.6373	0.1578

Table 5.1: Baseline simulations with five seeds and one million grains per size. Mean heights and topplings include all input events and 95% seed interval half-widths. Predictions come from the sparse systems. The last columns give zero-avalanche probability, zero-loss probability among avalanches, and loss probability per input.

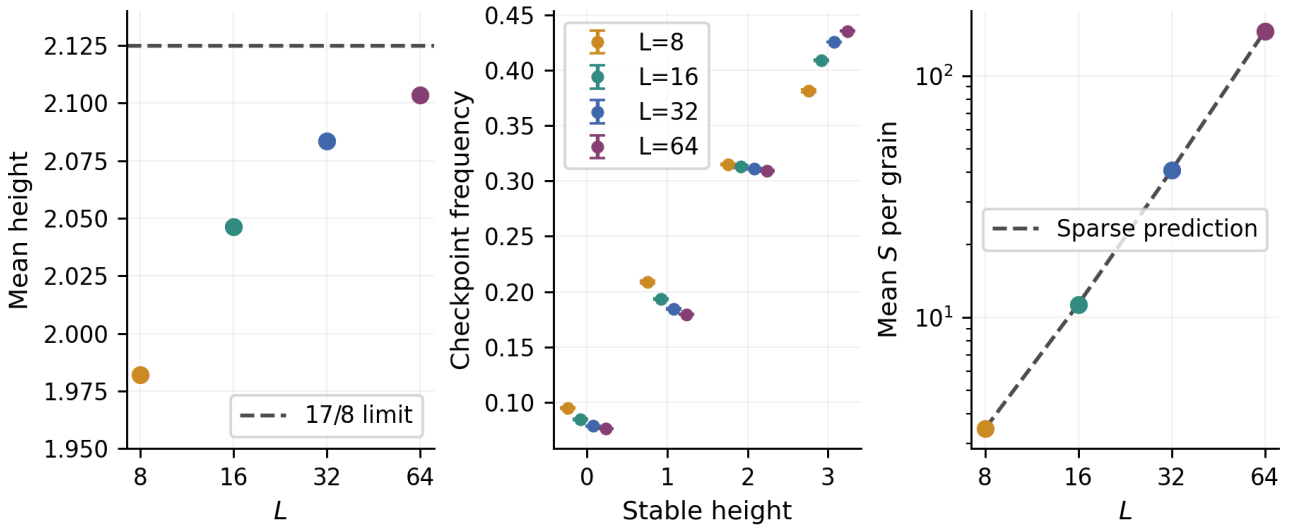


Figure 5.2: Stable heights have unequal frequencies, mean height increases with L, and measured mean topplings agree with finite-grid predictions. Each size uses one million measured inputs across five seeds. Left: mean height and the 17/8 infinite-volume reference. Centre: height frequencies after stabilization at regular intervals. Right: all-input mean topplings and independent predictions. Error bars are 95% seed intervals; some are smaller than the markers.

At $L = 64$, mean topplings are 153.033, with interval $[152.606, 153.459]$ containing the prediction 153.043. All four baseline sizes agree with their finite-grid benchmarks at the reported precision. Mean loss per grain is 0.9998, with interval $[0.9996, 1.0000]$; the endpoint mass accounts exactly for its small departure from one. As L increases, loss occurs less often but dissipative events lose more grains: $\mathbb{E}[D \mid D > 0]$ rises from 3.426 to 6.337. Larger grids carry roughly the same mean flux through rarer, larger loss events.

5.1.3 Time versus independent ensemble

For $L = 32$, one long trajectory gives time-sampled mean height 2.08420, interval $[2.08358, 2.08482]$, and height-three fraction 0.42566, interval $[0.42520, 0.42613]$. Sampling 100 independent chains after $20L^2$ inputs gives 2.08451, interval $[2.08182, 2.08720]$, and 0.42676, interval $[0.42445, 0.42906]$, respectively. The intervals overlap. At the earlier ages of L^2 and $2L^2$ inputs, the ensemble summaries still show the effect of starting empty.

The time and ensemble estimates remain consistent when the block length changes. Height has persistent positive dependence, whereas topplings show short-lag anticorrelation, so their uncertainty requires separate dependence diagnostics.

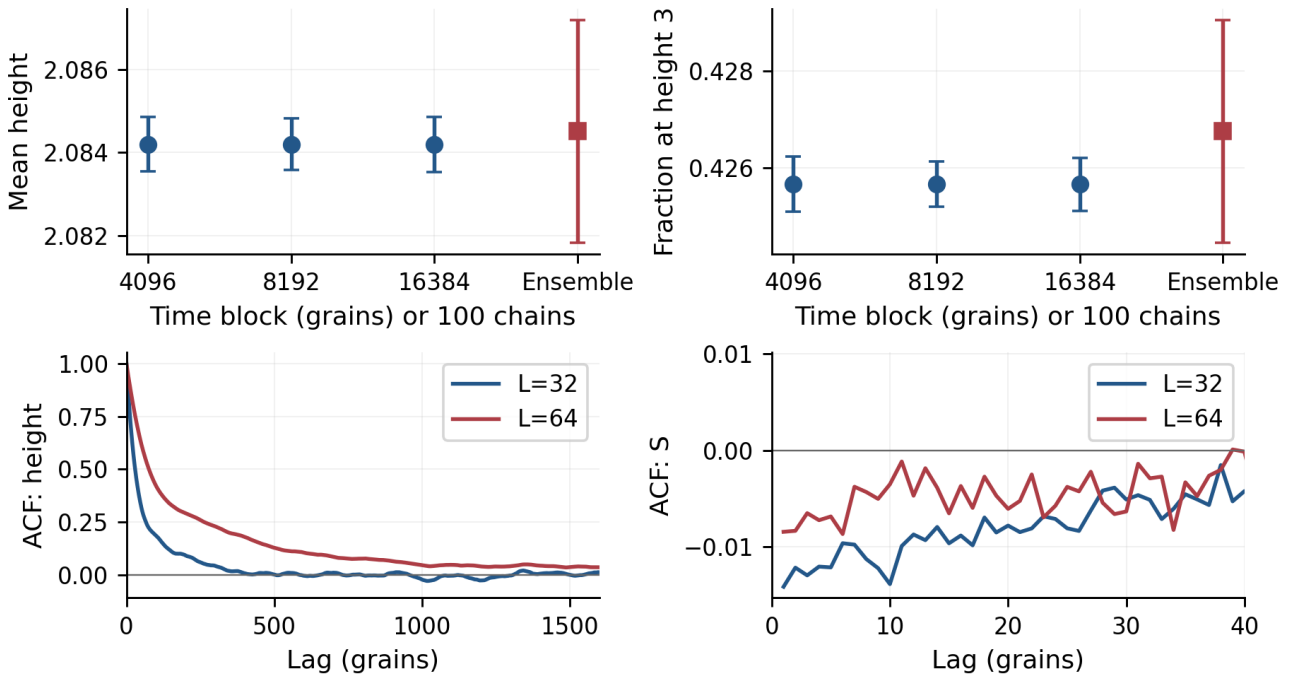


Figure 5.3: Time and ensemble estimates agree within their uncertainty. Top: estimates and 95% intervals from one $L = 32$ trajectory at three block lengths, compared with 100 independent chains at age $20L^2$. Time estimates cover 196,608 grains, with height-three frequency sampled every 128 grains. Bottom: height and toppling autocorrelation functions (ACFs) for the first run at $L = 32, 64$. The ensemble unit is an independent chain.

5.2 Avalanche distributions and finite-range fits

At $L = 64$, the largest observed toppling count is 19687, above the fitting cap $L^2 = 4096$. These larger events remain in the means and empirical distributions (Figure 5.4). The bounds L^2 for area and $2L - 1$ for single-source open reach reflect the finite grid directly. Loss measures transport to the boundary and differs in distribution from topples or area.

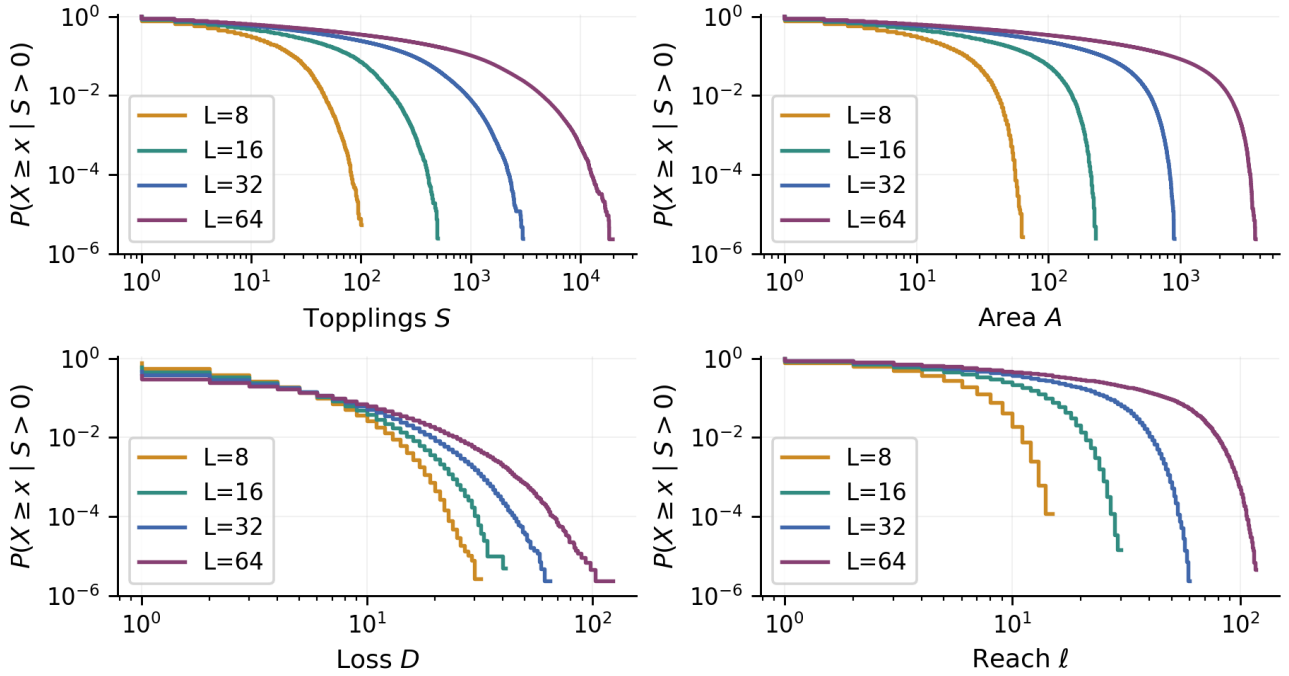


Figure 5.4: Avalanches extend to larger sizes as L grows, with different shapes for topplings, area, loss and reach. Baseline CCDFs use $L = 8, 16, 32, 64$, five seeds and one million inputs per size, conditional on $S > 0$. Left-continuous steps show inclusive $P(X \geq x)$. Zero loss remains in the normalization but is absent on log axes; Table 5.1 gives its mass.

Zero events make up a substantial part of the sample. At $L = 64$, $P(S = 0) = 0.5649$, and $P(D = 0 \mid S > 0) = 0.6373$. A positive-loss tail describes the dissipative subset of avalanches.

Quantity: support	Model	Retained n	α	λ	KS
S: [16,4096]	Power	245,477	1.1426 ± 0.0024	–	0.0310
	Cutoff	245,477	1.0090 ± 0.0082	0.000312 ± 0.000016	0.0074
	Exp.	245,477	–	0.002109 ± 0.000011	0.2468
A: [8,4096]	Power	291,321	1.1354 ± 0.0017	–	0.0461
	Cutoff	291,321	0.9591 ± 0.0023	0.000537 ± 0.000007	0.0185
	Exp.	291,321	–	0.002783 ± 0.000018	0.2717
D: [1,12289]	Power	157,789	1.5098 ± 0.0031	–	0.2242
	Cutoff	157,789	0.2627 ± 0.0312	0.1365 ± 0.0037	0.0247
	Exp.	157,789	–	0.1718 ± 0.0009	0.0468
ℓ: [3,127]	Power	345,680	1.2171 ± 0.0039	–	0.0713
	Cutoff	345,680	0.5817 ± 0.0040	0.0293 ± 0.0002	0.0260
	Exp.	345,680	–	0.0529 ± 0.0003	0.0895

Table 5.2: Discrete marginal fits at $L = 64$, using identical support for the three models within each observable. Parameters are means ± 95% half-widths across five independent seed fits; n is the total retained count, and KS is the mean descriptive CDF discrepancy. Power fixes $\lambda = 0$ and exponential fixes $\alpha = 0$. The S cap selects an intermediate range; other caps are physical bounds. Loss fits condition on $D > 0$; the other fits use their displayed positive intervals.

On $16 \leq S \leq 4096$, the pure-power effective exponent is 1.1426, with interval $[1.1403, 1.1450]$ (Table 5.2). Adding an exponential cutoff reduces mean KS from 0.0310 to 0.0074; an exponential alone gives 0.2468. The cutoff model gains 0.0100 nats per retained observation in mean marginal log score over power. Its CDF follows the observed curvature more closely on this range (Figure 5.5).

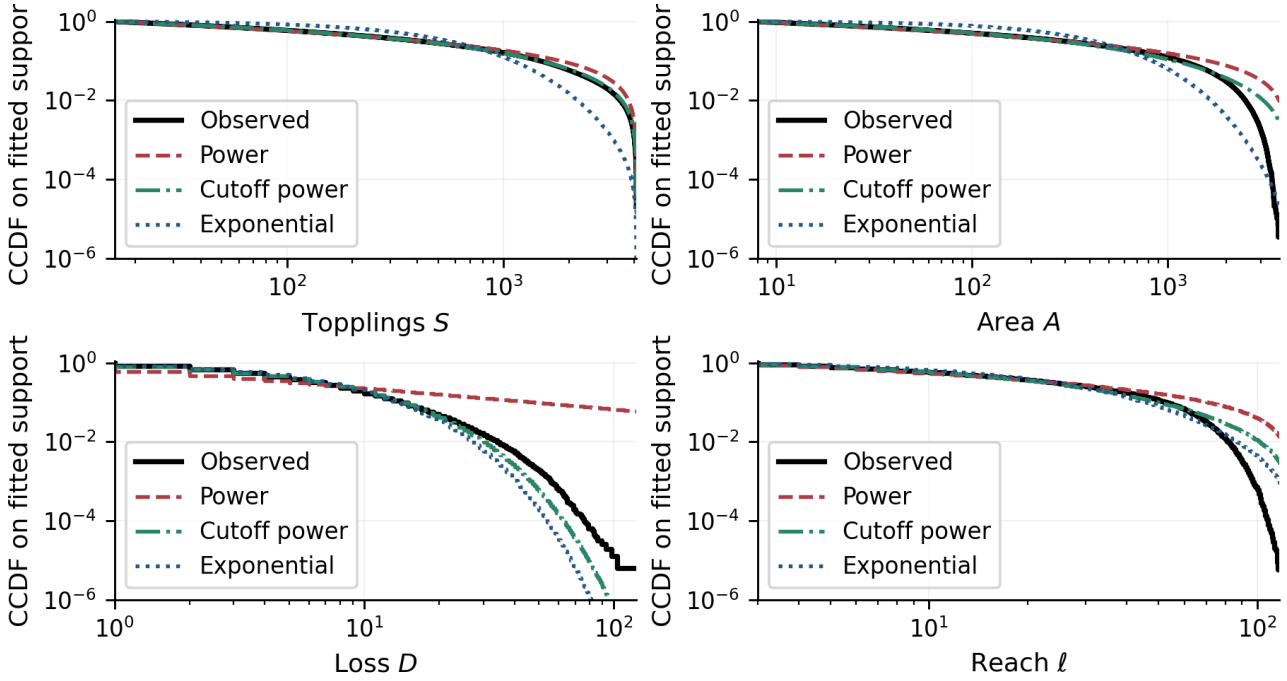


Figure 5.5: Observed and fitted CCDFs at $L = 64$, conditional on the intervals in Table 5.2. Observed curves pool the five equal-length runs. Model curves average the five separately normalized fitted PMFs before cumulation. Normalization uses the full fitting support, even where the display stops near the observed maximum. Cutoff models follow the curves more closely, with some systematic residual shape.

Area also shows curvature: the power and cutoff KS distances are 0.0461 and 0.0185. For positive loss, a pure power gives KS 0.2242, compared with 0.0468 for an exponential and 0.0247 for cutoff power. Reach has a curved finite-support distribution too.

The estimated slopes depend on the fitting range. At $L = 64$, halving and doubling the chosen lower limits gives pure-power estimates spanning 1.122–1.169 for S, 1.112–1.163 for area, 1.510–1.803 for positive loss and 1.008–1.366 for reach. This variation is separate from the narrower uncertainty intervals conditional on one range. The fitted exponents therefore describe the finite ranges examined here. Earlier work also found that simple scaling fails for BTW toppling statistics [4].

An additional upper-limit check uses the same five $L = 64$ runs and $x_{\min} = 16$, with $x_{\max} = 2048, 4096, 8192$. The retained counts are 231,863, 245,477, 250,246, respectively. Mean cutoff KS distances are 0.0027, 0.0074, 0.0097, compared with 0.0090, 0.0310, 0.0537 for power, and pure-power

exponents span 1.092–1.203. Curvature is preferred descriptively at all three caps, while the parameter estimates remain range-dependent.

5.3 Correlations and the role of the source position

We expected a positive correlation between S and A: every site counted in the area must topple at least once, and some sites topple repeatedly. Larger areas should also tend to have greater spatial extent. Loss should depend on these quantities and on the input site's distance to a dissipative boundary. For a single-grain input at (i_t, j_t) on the open square, we use

$$b_t = \min(i_t, j_t, L - 1 - i_t, L - 1 - j_t).$$

This is the distance to the nearest dissipative boundary site, with $b_t = 0$ on the boundary. Figure 5.6 compares loss probabilities within fixed size bands and b groups. All six Spearman correlations use $S > 0$ and retain zero-loss avalanches.

At $L = 64$, $\rho(S, A) = 0.999188$ with interval $[0.999175, 0.999201]$, $\rho(S, \ell) = 0.9879$ with $[0.9878, 0.9881]$, and $\rho(A, \ell) = 0.9908$ with $[0.9907, 0.9909]$. Repeated topplings and differences in cluster shape allow these observables to differ even when their ranks agree closely. Loss is less strongly correlated: $\rho(S, D) = 0.5349$ with $[0.5321, 0.5377]$, $\rho(A, D) = 0.5332$ with $[0.5304, 0.5360]$, and $\rho(D, \ell) = 0.5237$ with $[0.5206, 0.5267]$.

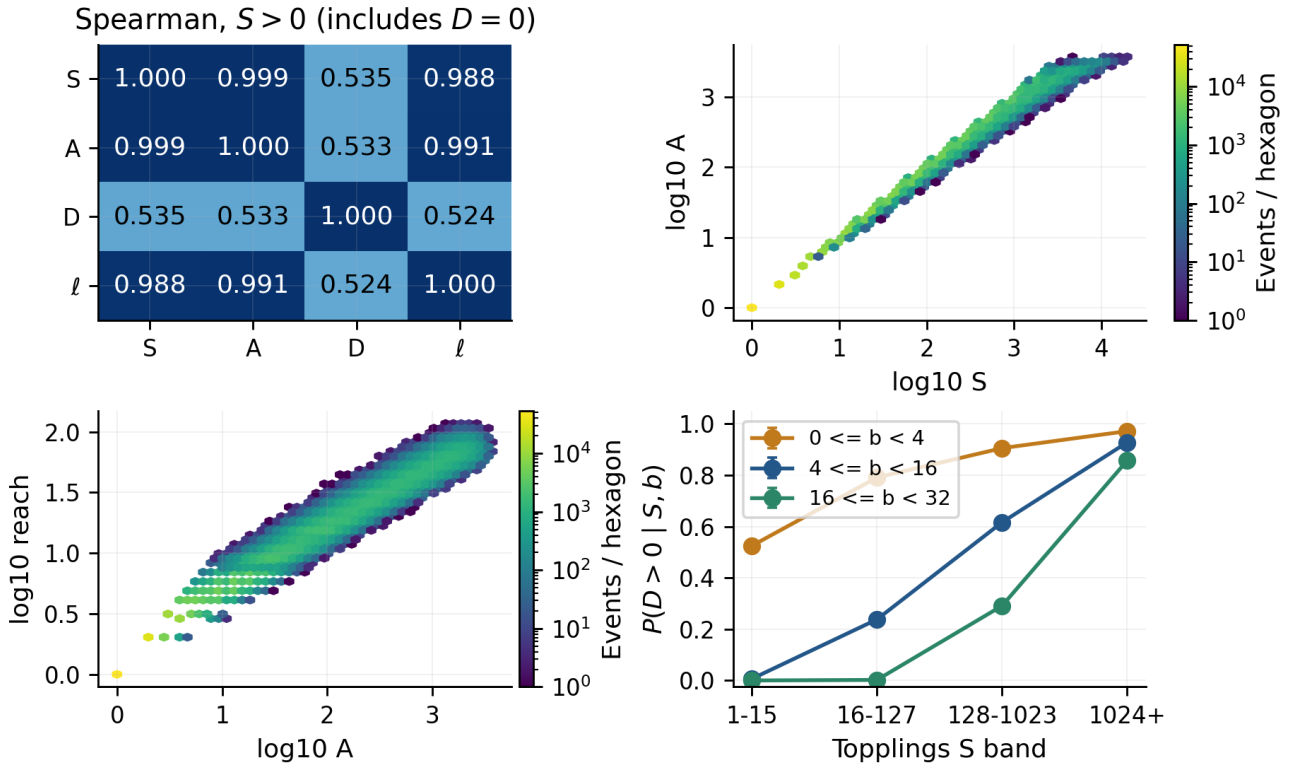


Figure 5.6: Size, area and reach track one another closely; loss depends more on sink access. Baseline $L = 64$ uses 435,058 nonzero avalanches across five seeds, retaining $D = 0$. Top left: mean seed Spearman matrix. Top right and bottom left: joint log-coordinate counts for the full nonzero sample; colour gives events per hexagon. Bottom right: loss probability within fixed size bands and source boundary-distance b strata, with 95% seed intervals.

Within the plotted size bands, avalanches starting nearer a sink have higher dissipation probabilities. Activity from an interior source must reach a boundary before any grain can leave, which helps explain the weaker association between loss and topplings. The grouping compares events of similar size; other avalanche features can still vary within each group. Each grid size is analysed separately.

5.4 Controlled changes to drive, boundaries and grouping

Table 5.3 compares each extension with its control at the same size. Open and cylindrical runs receive the same grain budget. Batch runs receive the same total grains in a quarter as many events. In each model, conservation gives stationary mean loss one per grain, while the measurements show how that loss is distributed across events.

L	Condition	Mean height	S/grain	$P(S>0)$	$P(D>0)$	$E[D \mid D>0]$
32	Open/uniform	2.08355 ± 0.00044	40.61 ± 0.07	0.4257	0.2024	4.941
32	Open/biased	2.08297 ± 0.00050	69.02 ± 0.06	0.4409	0.1396	7.166
32	Cylinder	2.10369 ± 0.00059	93.55 ± 0.12	0.4351	0.1687	5.927
32	Batch, $m=4$	2.08355 ± 0.00044	40.61 ± 0.07	0.8917	0.5983	6.686
64	Open/uniform	2.10361 ± 0.00025	153.03 ± 0.43	0.4351	0.1578	6.337
64	Open/biased	2.10313 ± 0.00048	266.67 ± 0.19	0.4442	0.1035	9.662
64	Cylinder	2.11401 ± 0.00053	357.51 ± 0.09	0.4406	0.1271	7.869
64	Batch, $m=4$	2.10361 ± 0.00025	153.03 ± 0.43	0.8978	0.4984	8.025

Table 5.3: Comparisons at matched sizes, with five seeds and one million measured grains per row. Mean height and topplings per grain include 95% seed half-widths. Probabilities and conditional loss use single-grain events for the first three rows at each L and four-grain events for batches. Batch mean loss is near four per event and one per grain. Batch estimates are paired with baseline.

5.4.1 Full-support spatial bias

On the open square, we bias the input towards the centre using

$$q_\eta(x) = \frac{1-\eta}{L^2} + \eta \frac{\exp[-\|x-c\|_2^2/(2\sigma^2)]}{\sum_y \exp[-\|y-c\|_2^2/(2\sigma^2)]}, \quad \eta = 0.9, \quad \sigma = L/8, \quad c = ((L-1)/2, (L-1)/2). \tag{5.1}$$

Every site has positive input probability, bounded below by $0.1/L^2$, so the recurrent stationary measure and its height statistics are preserved. The stationary mean response $\mathbb{E}[S] = q^\top w$ weights sites by their input probabilities. Bias towards central sites, farther from the sinks, therefore predicts more topplings per grain.

At $L = 64$, mean topplings rise from 153.033 to 266.669, close to the actual-q benchmark 266.627 (Figure 5.7). Dissipation probability falls from 0.1578 to 0.1035, while mean loss conditional on dissipation rises from 6.337 to 9.662. Mean loss per grain is 0.9999. The triggering probability can also change: biased input samples sites with different spatial weights even when the stationary law of the grid stays the same.

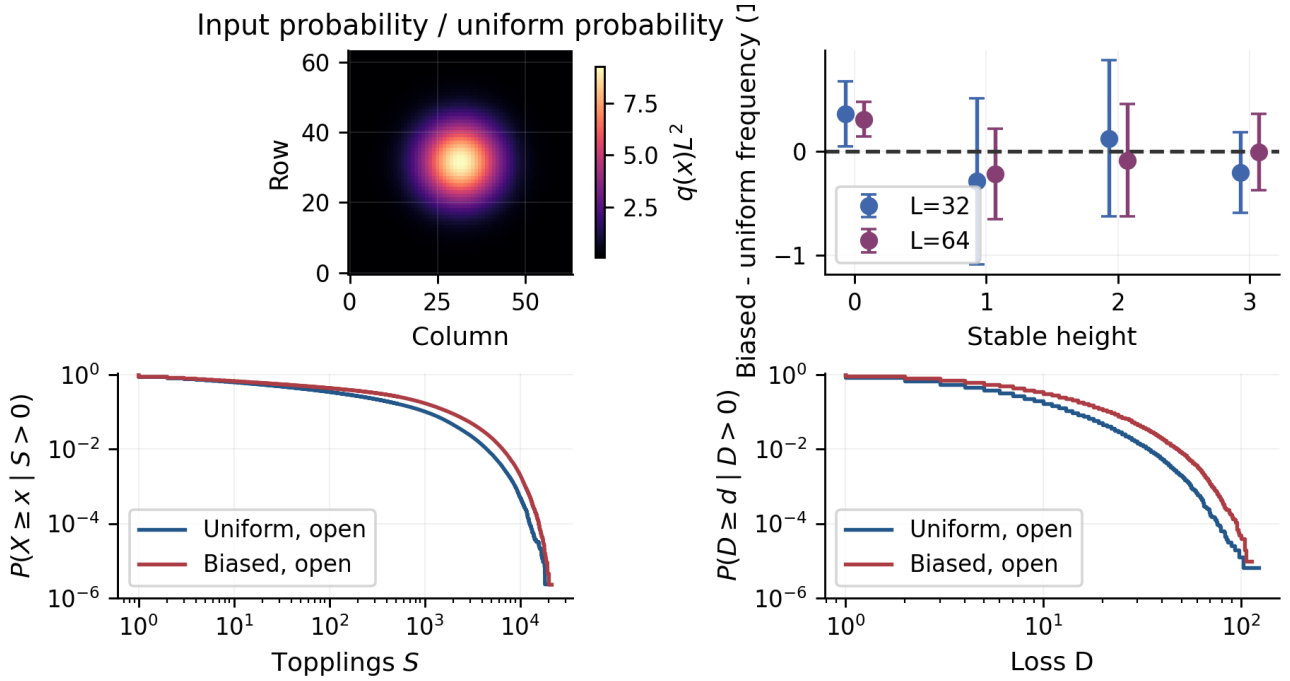


Figure 5.7: Spatial bias changes avalanche responses on the same graph. Top left: $L = 64$ input probability relative to uniform. Top right: biased-minus-uniform height-frequency differences in the primary runs at $L = 32, 64$, with Welch 95% independent-seed intervals; stationary theory predicts zero difference. Bottom: $L = 64$ size ($S > 0$) and loss ($D > 0$) CCDFs, with five seeds per condition.

The primary $L = 32$ height difference was -0.000578, with nominal interval $[-0.001132, -0.000024]$ narrowly excluding zero among several inspected contrasts. At $L = 64$ it was -0.000474, interval $[-0.000948, 0.000001]$. We investigated with five new seeds for each model at both sizes, giving

every condition the same longer budget: $100L^2$ burn-in and 800,000 measured grains per chain. The confirmation differences were 0.000196, interval $[-0.000265, 0.000658]$, and -0.000005, interval $[-0.000234, 0.000224]$, respectively. Both intervals contain zero; the longer runs did not reproduce the initial height deficit.

5.4.2 Cylindrical boundaries

For the cylinder, columns are periodic and grains leave only above the top row or below the bottom row. At the simulated widths $L \geq 3$, a toppling on a dissipative row loses one grain, including at the periodic seam. The graph distance is

$$d_{\mathrm{cyl}}((i,j),(i_0,j_0)) = |i - i_0| + \min(|j - j_0|, L - |j - j_0|), \tag{5.2}$$

and the source distance to a dissipative boundary is $\min(i, L-1-i)$. Reach follows this cylindrical distance and is bounded by $L + \lfloor L/2 \rfloor$.

A finite-grid benchmark for the stationary mean follows by taking w constant along each row. Set $w_{-1} = w_L = 0$ and

$$w(i,j) = \frac{(i+1)(L-i)}{2}. \tag{5.3}$$

The horizontal neighbours cancel two diagonal contributions, giving $(K_{\mathrm{cyl}} w)_{i,j} = 2w_i - w_{i-1} - w_{i+1} = 1$, including on the dissipative rows. Averaging over uniform input yields

$$\mathbb{E}[S]_{\mathrm{cyl}} = \frac{1}{L} \sum_{i=0}^{L-1} \frac{(i+1)(L-i)}{2} = \frac{(L+1)(L+2)}{12}. \tag{5.4}$$

Substitution into the sparse matrix verifies this analytic result.

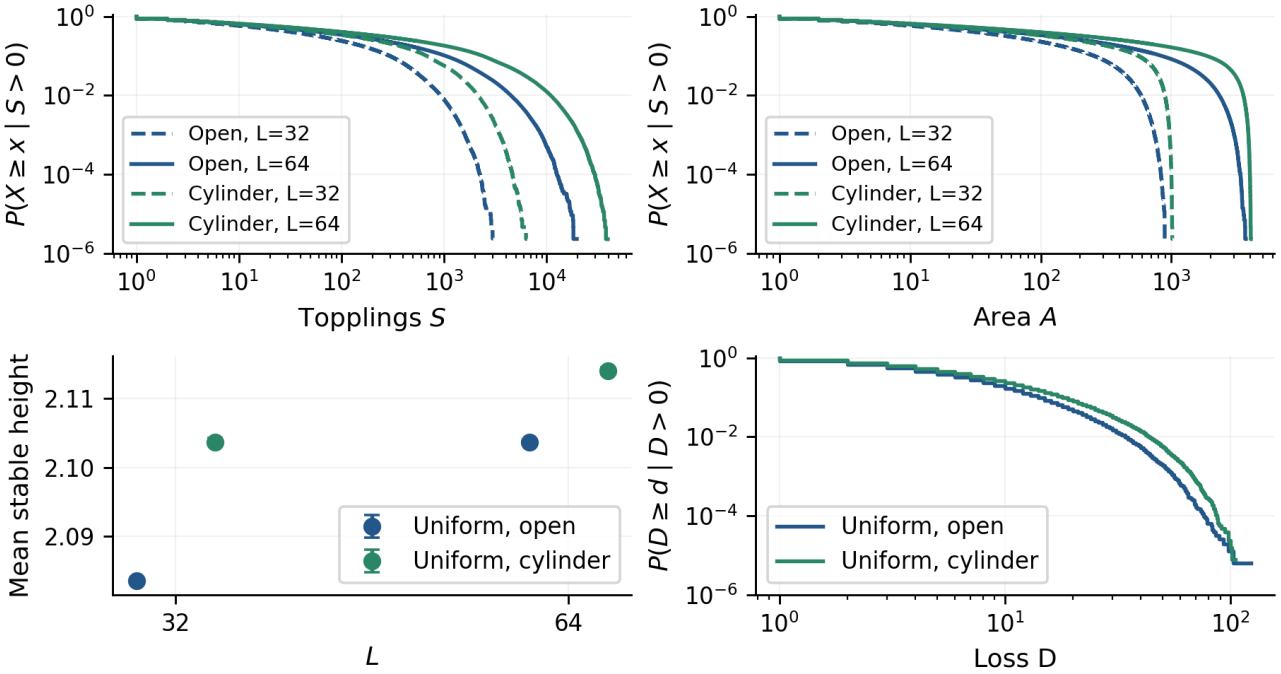


Figure 5.8: Cylindrical boundaries increase mean height and change avalanche size and loss distributions. Top: size and area CCDFs at $L = 32, 64$, conditional on $S > 0$. Bottom left: mean stable height with 95% seed intervals. Bottom right: positive-loss ($D > 0$) CCDFs at $L = 64$. Each curve uses five primary seeds and one grain per event.

Equation (5.4) predicts 93.5 and 357.5 topplings at $L = 32, 64$. The measured means are 93.547 and 357.511; at $L = 64$, the interval $[357.416, 357.605]$ contains the benchmark. Mean height exceeds the open control by 0.010406, interval $[0.009887, 0.010925]$ (Figure 5.8). With fewer dissipative edges, grains stay longer and generate more topplings. The boundary change affects both height statistics and response, while stationary loss remains one per input grain. At $L = 64$, mean loss is 1.0000, with dissipation probability 0.1271 and conditional mean loss 7.869.

5.4.3 Batches as an exact grouping experiment

Each batch draws four independent uniform sites with replacement and adds their grains before relaxation. We compare this with sequential single-grain additions from the same stable h, using the same ordered sites $s_1, \ldots, s_m$. A toppling in the sequential schedule remains legal if later, nonnegative inputs are already present. Concatenating those schedules therefore stabilizes the batched input legally. Uniqueness of the odometer gives

$$h_{\text{batch}} = h_{\text{sequential}}, \quad u_{\text{batch}} = \sum_{r=1}^{m} u_r, \quad S_{\text{batch}} = \sum_r S_r, \quad D_{\text{batch}} = \sum_r D_r. \tag{5.5}$$

Area counts the union of toppled sites. The sum $\sum_r A_r$ counts shared sites more than once, whereas the toppling and loss identities hold even for correlated consecutive avalanches.

If P is the one-grain transition, independent sources give batch transition P^m. Every stationary measure of P is stationary for P^m. Depending on the sampling period, P^m can still have more than one recurrent class or fail to converge from a particular start. We use a matched baseline grid after equilibration so that the comparison isolates grouping. Each batch is fully relaxed before the next one arrives.

Across both sizes, all 500,000 batches match the grouped controls exactly in S, D and final mass. Grid configurations and height frequencies also agree whenever sampled. Independent small-grid tests verify the full odometer and area-union identities. At $L = 64$, 12.01% of batches have overlapping toppled sets, and summing sequential areas exceeds the mean union by 16.64 sites per batch. These overlaps account for the difference between the two area curves in Figure 5.9.

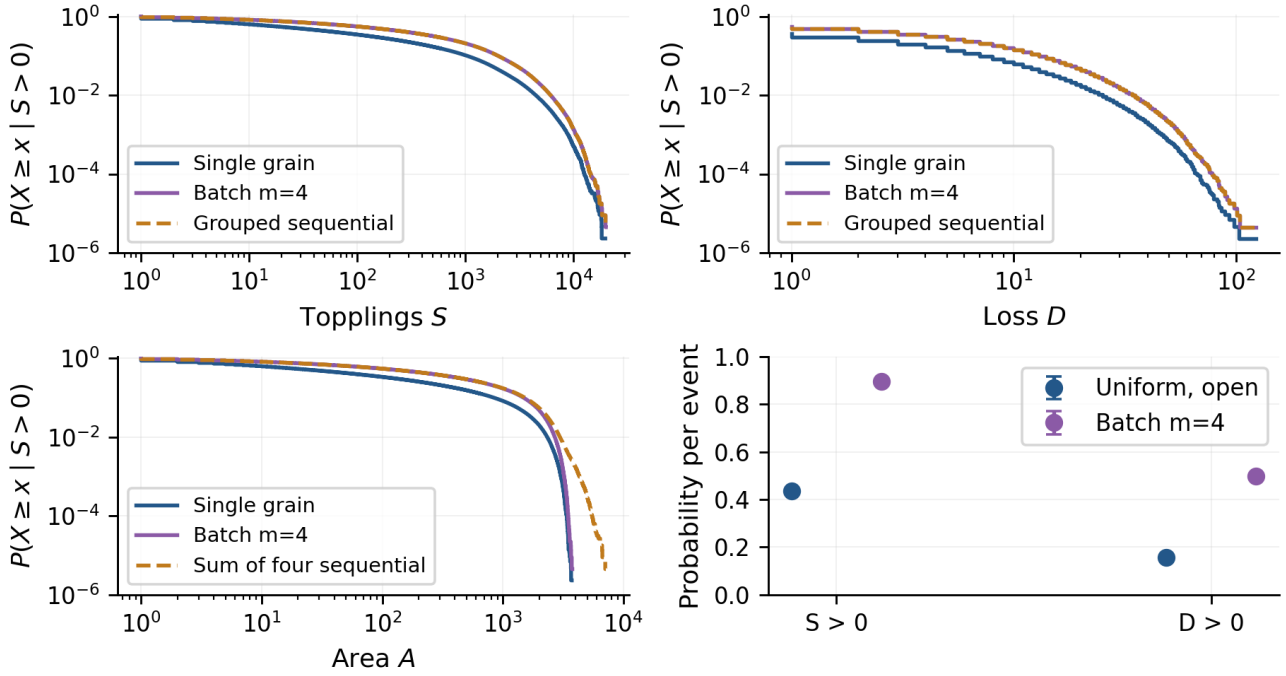


Figure 5.9: Event grouping at $L = 64$ for five matched runs: one million baseline single-grain events and 250,000 four-grain batches. Grouped sequential size and loss coincide with the batch values. For area, summing four sequential areas counts shared sites repeatedly; the batch counts their union. CCDFs condition on nonzero grouped or ungrouped avalanches as appropriate. Bottom right: unconditional triggering and dissipation probabilities with 95% seed intervals.

Grouping raises toppling probability per event from 0.4351 to 0.8978 and loss probability from 0.1578 to 0.4984. Mean loss is about four per batch, with exactly the baseline value per grain. The higher event probabilities reflect four successive inputs combined into one event.

Chapter 6

Discussion, limitations, and conclusion

A fixed local toppling rule produces broad avalanche responses as inputs encounter different configurations. The baseline height frequencies explain why the measured mean exceeds the equal-frequency estimate of $3/2$ and approaches the infinite-volume reference as the lattice grows. Mean topplings agree with the finite-grid benchmarks. Size, area and reach are strongly associated, while the source-distance comparison shows how access to a sink affects loss.

Central bias and cylindrical boundaries both increase mean topplings, through different mechanisms. Bias changes the weights assigned to input sites under an invariant recurrent measure; the independent confirmation found height differences compatible with zero. The cylinder changes the graph itself, producing higher mean heights and a response predicted by its analytic benchmark. In Eq. (3.8), these changes enter through q and K, respectively. Four-grain batches instead combine successive, potentially overlapping responses. Their exact agreement with grouped sequential totals shows how event distributions can change while per-grain topplings and loss remain fixed.

The statistical conclusions cover four baseline sizes, two sizes per extension and five primary seeds per condition. Seed and block estimates account for temporal dependence, with precision limited by the number of independent runs. Burn-in was assessed through the measured observables; convergence of the full configuration distribution was not established. Cutoff models describe the curvature better over the fitted ranges, while sensitivity to those ranges and residual shape limit the interpretation of their parameters. The reported speed gains apply to the specified timing workload.

Across the comparisons, conservation fixes long-run mean loss at one per input grain. Larger lattices carry this flux through rarer events with greater loss, and batch grouping raises loss per event in proportion to its input. The combination of exact balance relations and event distributions explains how similar average transport can coexist with markedly different avalanche statistics.

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